

Python Assignment 1

Data Structures – Strings & Tuples

This assignment covers basic string operations (concatenation, slicing, methods) and tuple operations (creation, modification, access) in Python.

Strings

1. String Concatenation

Join a greeting, a user-entered name and a welcome message into one sentence.

```
# 1. String Concatenation:

Text = "Hello "
name = input("Enter your name:")
Welcome = ("Welcome to Python programming")

print(Text + name + "," + Welcome)
```

Output

```
Enter your name:Revathi
Hello Revathi,Welcome to Python programming
```

2. String Slicing and Indexing

```
# 2. String Slicing and Indexing:

text = "Hello Revathi,Welcome to Python programming"

# a. Print the first character of the string.
print(text[0])

# b. Print the last character of the string.
print(text[42])
print(text[-1])

# c. Print the first 5 characters of the string.
print(text[0:5])

# d. Print the last 11 characters of the string.
print(text[-11:])

# e. Print the string in reverse.
print(text[::-1])

# f. Use slicing and print the word "Python" from the existing string.
print(text[-18:-12])
print(text[25:31])
```

Output

```
H
```

```
g
g
Hello
programming
gnimmargorp nohtyP ot emocleW,ihtaveR olleH
Python
Python
```

Notes

Expression	Meaning
text[0]	First character (index starts at 0)
text[-1]	Last character (negative index counts from the end)
text[0:5]	Characters from index 0 up to (not including) 5
text[-11:]	Last 11 characters
text[::-1]	Whole string reversed (step of -1)

3. String Methods

```
# 3. String Methods:

strM = "Python beginner tutorial"

# a. Convert the sentence to uppercase.
print(strM.upper())

# b. Convert the sentence to lowercase.
print(strM.lower())

# c. Use capitalize and return the sentence to the original input form.
str = "PYTHON BEGINNER TUTORIAL"
print(str.capitalize())

# d. Count the total number of occurrences of character 't' in the string.
print(strM.count("t"))

# e. Replace all occurrences of "Python" with "Data Analytics"
strM = "Python beginner tutorial"
print(strM.replace("Python", "Data Analytics"))
```

Output

```
PYTHON BEGINNER TUTORIAL
python beginner tutorial
Python beginner tutorial
3
Data Analytics beginner tutorial
```

Tuples

Creation, modification (by building new tuples) and access.

```
# Tuples (Creation, Modification and Access):
```

```
tuple1 = 10, 20, 30
tuple2 = 40, 50, 60

# a. Concatenate the two tuples and store it in "t_combine"
t_combine = tuple1 + tuple2
print(t_combine)

# b. Repeat the elements of "t_combine" 3 times
print(t_combine * 3)

# c. Access the 3rd element from "t_combine"
print(t_combine[2])

# d. Access the first three elements from "t_combine"
print(t_combine[0:3])

# e. Access the last three elements from "t_combine"
print(t_combine[-3:])
```

Output

```
(10, 20, 30, 40, 50, 60)
(10, 20, 30, 40, 50, 60, 10, 20, 30, 40, 50, 60, 10, 20, 30, 40, 50, 60)
30
(10, 20, 30)
(40, 50, 60)
```

Concepts Covered

- String concatenation with +
- Indexing (positive and negative) and slicing [start:stop:step]
- String methods: upper(), lower(), capitalize(), count(), replace()
- Tuple concatenation (+), repetition (*), indexing and slicing
- Tuples are immutable, so "modifying" one means creating a new tuple

How to Run

```
python assignment1.py
```

~~Requires Python 3.x. No external libraries needed.~~